Predicting Brain Tumor Type Based on MRI Images

DS 4002: 04/20/26

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Outline:

Project Details

Data

Analysis Plan and Justification

Analysis Decisions and Bias and Uncertainty Validation

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Project Motivation and Research Design

Motivation:
Early diagnosis of brain tumors can improve patient outcomes. Because MRI scans contain important visual information, we tested whether a machine learning model could classify tumor type directly from the image [1].

Research Question:
Can a machine learning model accurately classify MRI images as glioma, meningioma, pituitary tumor, or no tumor using only the image?

Hypothesis:
Our model will predict the diagnosis of the different tumor types with at least 80% classification accuracy.

Modeling Approach:

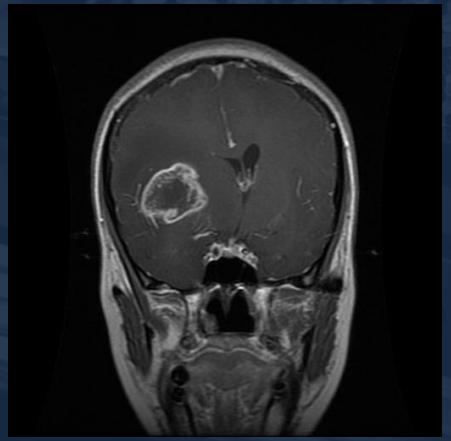
- Supervised multi-class CNN with MobileNetV2 transfer learning
- Resized and normalized MRI images used as model input
- Data augmentation used during training

Our Data

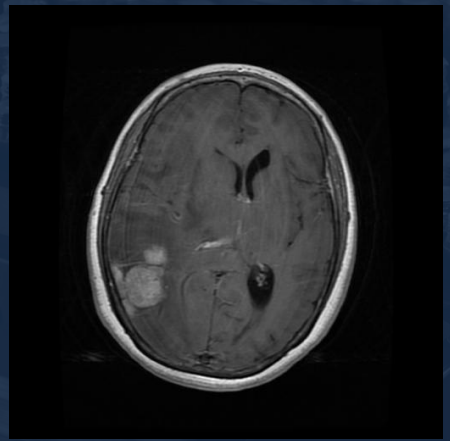
Categories	Description	Potential Responses
Glioma	A MRI image of a glioma tumor	A MRI image
Meningioma	A MRI image of a meningioma tumor	A MRI image
No tumor	A MRI image of no tumors	A MRI image
Pituitary	A MRI image of a pituitary tumor	A MRI image

- Acquisition: Mendeley Data [2]
- Format: .zip file containing .jpg images
- Structure:
  - Training Set
    - Glioma - 1300
    - Meningioma - 1300
    - No Tumor - 1600
    - Pituitary - 1400
  - Testing Set
    - Glioma - 300
    - Meningioma - 300
    - No Tumor - 405
    - Pituitary - 300

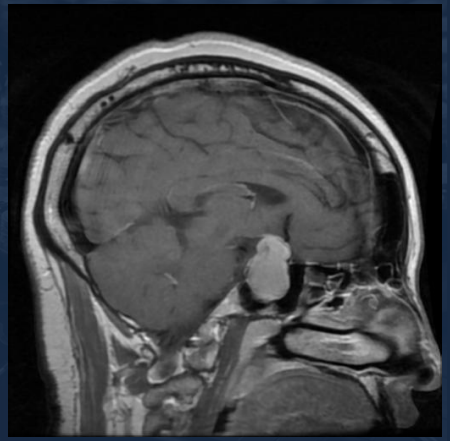
Examples of Each Image Type



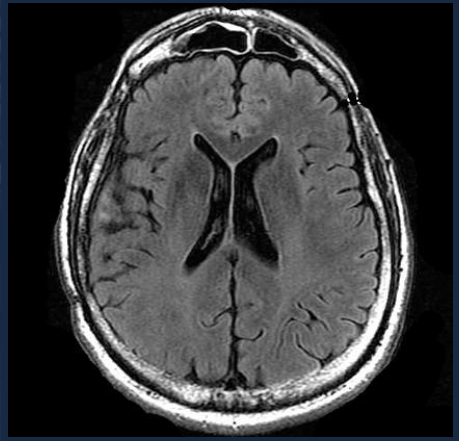
Glioma



Meningioma



Pituitary



No Tumor

Analysis Plan and Justification

Preprocessing: Clean and prepare data

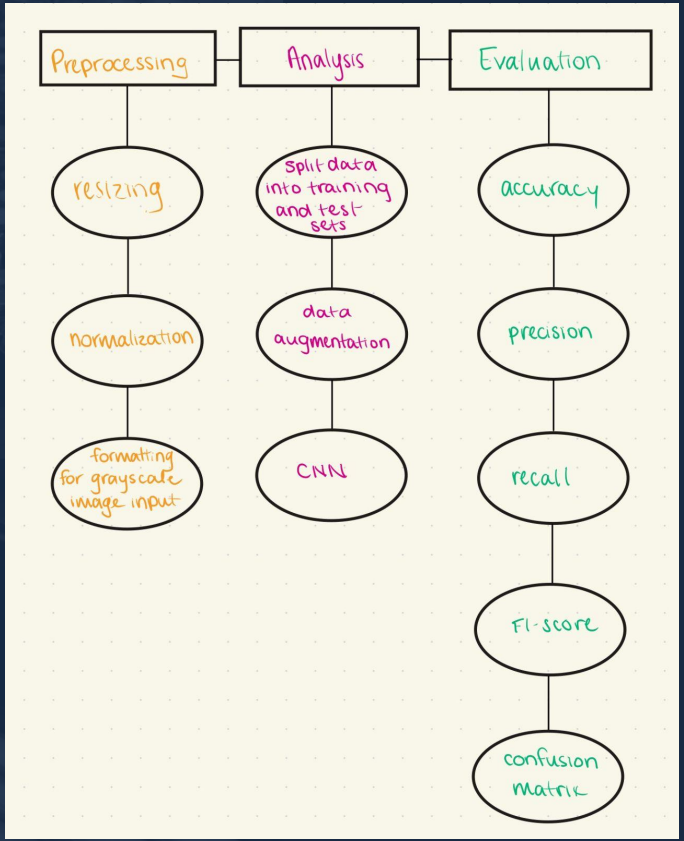
Observation: MRI images vary in size and formatting.
Decision: Resize images to consistent dimensions and normalize pixel values.

Analysis: Convolutional Neural Network: MobleNetV2

Observation: MRI scans contain spatial visual patterns, so a CNN is appropriate for learning differences among tumor classes.
Decision: Use MobileNetV2 transfer learning to improve feature extraction and training efficiency.

Evaluation: Model Performance

Observation: We want to measure both overall accuracy and where the model makes classification errors across classes.
Decision: Evaluate the model using accuracy, precision, recall, F1-score, and a confusion matrix.



Tricky Analysis Decisions

Keep the full four-class classification problem rather than simplify the task to tumor vs. no tumor.

A binary model would only tell us whether a scan showed a tumor or not

Our actual research question asked whether MRI images could distinguish among glioma, meningioma, pituitary tumor, and no tumor

Keeping all four classes preserved the clinical meaning of the project

However, multi-class classification is harder because some tumor types have overlapping visual features

Why it was tricky

- Binary classification would be easier and likely more accurate
- But it would reduce the usefulness of the results

Impact

- Preserved the full research goal
- Achieved over 80% accuracy
- Revealed meningioma as the hardest class to classify

Bias and Uncertainty Validation

Different Image Angles

- Images were taken at a variety of cross-section angles in each set
- Different angles may not be balanced within each target variable

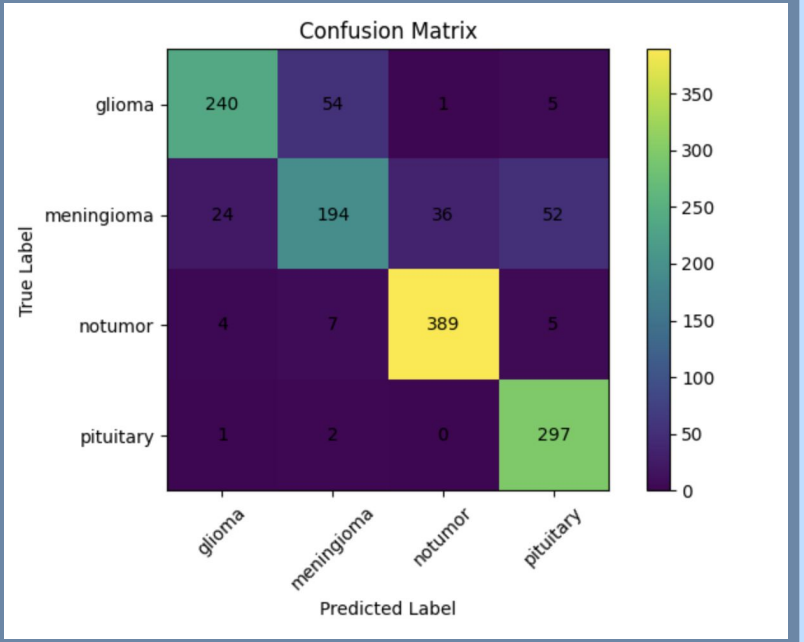
Class Representation and Similarity

- Some classes were still more difficult to distinguish
- Tumor types with overlapping visual patterns increase the chance of misclassification

How We Measured Uncertainty

- We assessed uncertainty using class-level precision, recall, and F1-score, and a confusion matrix
- These results showed that strong overall accuracy still included uncertainty in meningioma predictions

Results and Conclusions



	precision	recall	f1-score
glioma	0.89	0.80	0.84
meningioma	0.75	0.63	0.69
notumor	0.91	0.96	0.94
pituitary	0.83	0.99	0.90
accuracy			0.85
macro avg	0.85	0.85	0.84
weighted avg	0.85	0.85	0.85

- Best overall performance in no tumor class
  - Detects most pituitary tumors (but some false positives)
- Worst performance in meningioma class
  - Meningioma likely has overlapping features with other tumors

The hypothesis was mostly supported:
Overall model accuracy is > 80% (85%), but F1-score for meningioma is only 69%.

Next steps

- Expand this model to different types of cancer
- Expand this model to encompass stroke patients to analyze which patients have a blocked vessel vs a bleeding vessel for greater efficiency

NEW LINES OF EXPLORATION

- Improve performance in meningioma classification
- Add a classification for tumor size

POTENTIAL IMPROVEMENTS

- Does one imaging angle provide particularly better or worse diagnostic outcomes?
- Why is meningioma detection worse than the other classes?

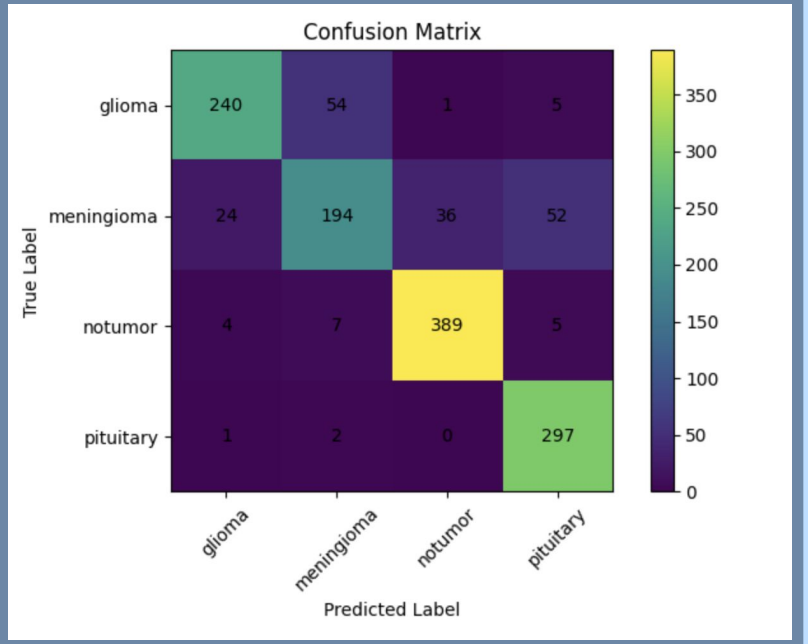
NEW QUESTIONS

References

[1] "Meningioma," *Brain Tumor Center*, 2023. https://med.stanford.edu/brain-tumor/conditions/meningioma.html
[2] A. Ghaffar, "Brain Tumor Data," *data.mendeley.com*, vol. 1, Jan. 2024, doi: https://doi.org/10.17632/w4sw3s9f59.1.

https://github.com/kaleigh-west/DS4002_Project3

Final Takeaways



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